

Question: 1

Draw a detailed network topology diagram for a small co-working space with 3 departments (e.g., Marketing, Tech, Finance), showing routers, switches, wireless access points, and user devices using any diagram tool like draw.io or Lucidchart?

Answer: • 1. Core Internet Gateway (ISP & Router):

- ISP (Internet Service Provider): External high-speed fiber internet source jo co-working space ko web access deta hai.

- Main Edge Router: Yeh pure office ka central gateway hota hai jo incoming and outgoing packets ko control karta hai, firewall security lagata hai aur unique IP addresses allocate karne ke liye DHCP manage karta hai.

- 2. Core Switch (Central Layer 3 Switch):

- Role: Yeh main router se connect hota hai aur poori building ke primary traffic ko separate departments me distribute karta hai.

- VLAN Configuration: Network security aur performance badhane ke liye is central switch par teen alag-alag VLANs (Virtual Local Area Networks) configure kiye jayenge taaki departments ka data aapas me mix na ho (e.g., VLAN 10 for Marketing, VLAN 20 for Tech, VLAN 30 for Finance).

- 3. Department Switches (Layer 2 Switches):

- Marketing Switch: Marketing team ke heavy media uploads, designs aur social dashboards handle karne wale high-speed desktop/wired systems ko local connectivity deta hai.

- Tech Switch: Tech department ke developers, servers aur local lab testing devices ke massive data transfers aur low latency requirements ko handle karta hai.

- Finance Switch: Yeh critical database systems, accounting software platforms aur billing hardware workstations ko highly isolated aur secure local access connectivity data hai.

- 4. Wireless Access Points (APs):

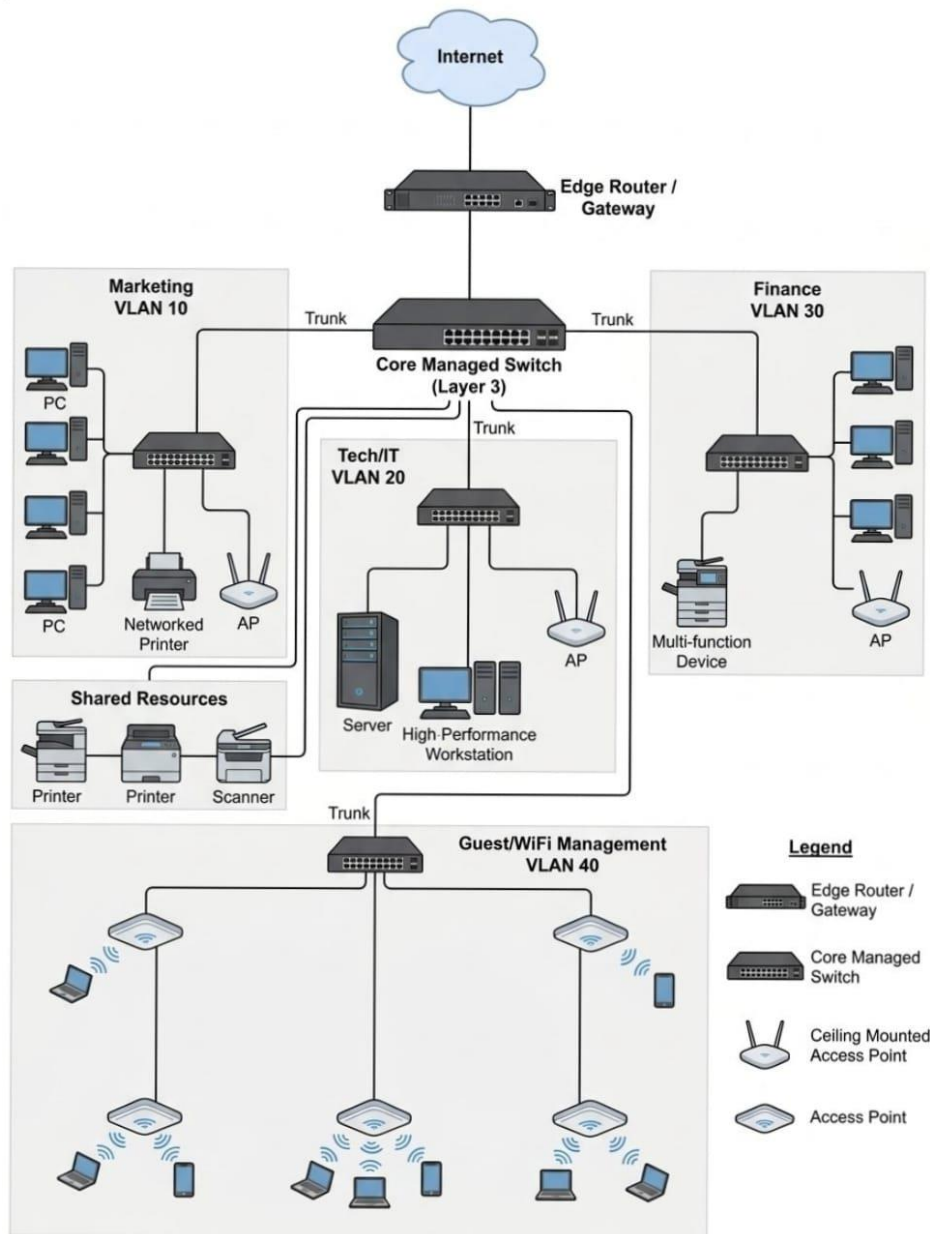
- Har department ke local zones ya common seating tables par ek dedicated Wireless Access Point (AP) setup kiya jata hai.

- Yeh corporate laptops, tablets aur employee smartphones ko secure dual-band Wi-Fi connection provide karta hai.

- 5. User Devices (End Nodes):

- Wired Devices: Production machines, main development systems, aur shared office network printers direct LAN Ethernet cables ke zariye department switches se jude hote hain.

- Wireless Devices: Mobile phones, remote worker laptops aur visiting clients ke devices dedicated secure Wi-Fi access points ke zariye connected hote hain.



Question: 2

Create an IP addressing spreadsheet for the above co-working space, assigning unique subnets and VLAN IDs to each department, and listing at least 5 device IPs per department.

Hint: Use private IP ranges (e.g., 192.168.x.x) and make sure there's no overlap between subnets?

1. Answer: Marketing Department (VLAN 10) • Subnet Range: 192.168.10.0/24 • Subnet Mask: 255.255.255.0 • Default Gateway: 192.168.10.1 (Router Interface) • 5 Device IP Allocations: • 192.168.10.10 - Department Network Printer • 192.168.10.21 - Designer Workstation PC 01 • 192.168.10.22 - Marketing Head Laptop • 192.168.10.23 - Content Creator iMac • 192.168.10.50 - Wireless Access Point (AP-Marketing)
2. Tech / IT Department (VLAN 20) • Subnet Range: 192.168.20.0/24 • Subnet Mask: 255.255.255.0 • Default Gateway: 192.168.20.1 (Router Interface) • 5 Device IP Allocations: • 192.168.20.10 - Local Testing Web Server • 192.168.20.21 - Developer Linux Workstation 01 • 192.168.20.22 - Developer Linux Workstation 02 • 192.168.20.23 - Network Admin Laptop • 192.168.20.50 - Wireless Access Point (AP-Tech)
3. Finance Department (VLAN 30) • Subnet Range: 192.168.30.0/24 • Subnet Mask: 255.255.255.0 • Default Gateway: 192.168.30.1 (Router Interface) • 5 Device IP Allocations: • 192.168.30.10 - Multi-function Secure Device (Printer/Scanner) • 192.168.30.21 - Accounting Ledger PC 01 • 192.168.30.22 - Payroll Processing PC • 192.168.30.23 - Finance Director Laptop • 192.168.30.50 - Wireless Access Point (AP-Finance)

Question: 3

Design a basic redundancy plan for the network that ensures internet connectivity is available even if one main switch fails. Briefly describe your approach and update your topology diagram to include redundancy?

Answer: • 1. Dual-WAN Router Configuration (Gateway Redundancy):

- Approach: Network me do routers setup kiye jayenge—ek Primary Router (jo main high-speed fiber connection handle karega) aur dusra Backup/Secondary Router (jo ek alternate ISP line se juda hoga).
- Protocol: Isme VRRP (Virtual Router Redundancy Protocol) ya HSRP configure kiya jayega. Dono routers ek single Virtual IP share karenge. Agar primary router band

hota hai, toh backup router bina users ko pta chale automatic internet connection handle kar lega.

- 2. Core Switch Redundancy (Switch Failures):

- Approach: Single L3 Core Switch ki jagah do interconnected Core Switches (Switch A aur Switch B) lagaye jayenge.

- Stacking/Interlink: Dono switches aapas me high-speed Trunk links (Aggregation Link) ke zariye jude honge. Agar Switch A purani architecture me crash ho jata tha toh poora office down ho jata tha, par ab Switch B instantly saare active VLAN lines ko control kar lega.

- 3. Redundant Cable Pathways (Link Redundancy):

- Approach: Har peripheral/department switch (Marketing, Tech, Finance) ko dual connections diye jayenge—ek cable Core Switch A me jayegi aur dusri backup Core Switch B me.

- Protocol (Spanning Tree Protocol - STP): Network loop se bachne ke liye RSTP (Rapid Spanning Tree Protocol) active rahega. Normal days me backup line block rahegi, par jaise hi main switch ya link fail hoga, backup path mili-seconds ke andar automatic unblock ho jayega.

- 4. Power Redundancy (UPS Integration):

- Main rack me lagaye gaye dono routers aur switches dedicated Dual UPS (Uninterruptible Power Supply) lines se jude honge taaki short-term electricity cutout hone par bhi switches drop na hon.

Question: 4

Select suitable network hardware (router, switches, access points) for the co-working space scenario and create a Bill of Materials (BOM) table with device model, quantity, and estimated price (use online prices from Indian retailers like Amazon or Flipkart)?

Answer: • 1. Core Router / Security Gateway:

- Selected Model: TP-Link ER605 Multi-WAN Wired VPN Router
- Purpose: Yeh router load-balancing aur dual-ISP connection redundancy ko fully support karta hai.
- 2. Core Layer 3 Managed Switch:
- Selected Model: Cisco Business 250 Series (CBS250-24T-4G)
- Purpose: Teeno departments ke alag-alag VLANs (VLAN 10, 20, 30) ke beech secure routing setup karne ke liye backbone interface ka kaam karega.
- 3. Wireless Access Points (APs):
- Selected Model: Aruba Instant On AP11 Indoor Access Point
- Purpose: Har department zone me dual-band user devices aur wireless roaming devices ko heavy high-density network support dene ke liye use hoga.
- 4. Core Backbone & Interconnect Cables:
- Selected Model: Cat6 UTP High-Speed Copper Cable Bundle Box
- Purpose: Switches, structural routers, wired server systems aur networked printers ko interconnect karne ke liye full gigabit baseline speeds provide karega.

Question: 5

Draft a short security policy for the co-working space network covering at least: WiFi password rules, VLAN access restrictions, guest network setup, and basic firewall guidelines.

Hint: Think about how Zomato or Paytm might separate guest and staff WiFi in their offices?

Answer: • 1. Wi-Fi Password Rules (Access Control):

- Staff Wi-Fi: Employees aur official desks ke liye WPA3-Enterprise authentication ka use kiya jayega, jahan har user ke paas unka apna unique login username aur strong alphanumeric password hoga.

- Guest Wi-Fi: Isme ek common standard password rahega jo har hafte automatic reset hoga.

- 2. VLAN Access Restrictions (Data Isolation):

- Finance Isolation: Finance department (VLAN 30) ke network ko baaki sabhi networks se poori tarah isolate rakha jayega. Marketing ya Guest network ka koi bhi user Finance database ko access nahi kar sakta.

- Tech / IT Management: Sirf Tech department (VLAN 20) ke administrators ko hi main router aur switches ke management terminal (SSH/HTTPS) ko open karne ki permission hogi.

- 3. Guest Network Setup (Captive Portal):

- Separate Network Interface: Guest devices ko ek alag Guest VLAN (VLAN 40) me daala jayega, jisse wo office ke internal network printers ya local storage servers ko na dekh sakein.

- Captive Portal Login: Jaise hi koi guest Wi-Fi se judega, unhe ek web portal par redirect kiya jayega jahan mobile number enter karke OTP verify karna zaroori hoga. Iske sath hi guest devices ke internet connection ka session max 4 hours ke liye hi active rahega.

- 4. Basic Firewall Guidelines (Threat Prevention):

- Inbound Traffic Control: Main network edge firewall par default "Deny All" rule set kiya jayega, jisse bahar ke internet se aane wala koi bhi unsolicited request block ho sake.

- Content & URL Filtering: P2P file-sharing networks (Torrents), dangerous adult sites, aur unverified third-party scripts ko gateway level par block kiya jayega taaki illegal downloads na ho sakein.